

Thromboembolic Disease: Deep Vein Thrombosis & Pulmonary Embolism

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September 2026

Session Objectives

- **Definitions**
- **Predisposing factors and Clinical risk factors**
- **Deep Vein Thrombosis**
- **Pulmonary Embolism**
- **Diagnostic Strategies for DVT and PE**
- **Prevention of DVT and PE**
- **Anticoagulant Therapy**
- **Thrombolytic Therapy**
- **Other Treatment Approaches**

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Venous Thromboembolic Disease

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- Pulmonary embolism
- Deep vein thrombophlebitis



**BAGHDAD, Iraq
(CNN) -- David
Bloom, a prominent
NBC News journalist
who was covering
the war in Iraq, died
suddenly of a non-
combat ailment
while on duty.**



Heavy D

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- **Coroner:**
“Heavy D died of pulmonary embolism”



Serena

**“Serena Williams undergoes
emergency treatment for
pulmonary embolism”**

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Regis

**“Regis to undergo surgery to
remove a blood clot”**

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Al Roker

"Last week I was admitted to the hospital with a blood clot in my leg which sent some clots into my lungs"

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Remains a critical diagnosis in 2026

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Clinical Review & Education

JAMA | Review

Acute Pulmonary Embolism A Review

Yonathan Freund, MD, PhD; Fleur Cohen-Aubart, MD, PhD; Ben Bloom, MD, PhD

JAMA. 2022;328(13):1336-1345. doi:10.1001/jama.2022.16815

Clinical Review & Education

JAMA | Review

Diagnosis and Treatment of Lower Extremity Venous Thromboembolism A Review

Romain Chopard, MD, PhD; Ida Ehlers Albertsen, MD, PhD; Gregory Piazza, MD, MS

JAMA. 2020;324(17):1765-1776. doi:10.1001/jama.2020.17272

The NEW ENGLAND JOURNAL of MEDICINE

CLINICAL PRACTICE

Caren G. Solomon, M.D., M.P.H., Editor

Pulmonary Embolism

Susan R. Kahn, M.D., and Kerstin de Wit, M.B., Ch.B., M.D.

This Journal feature begins with a case vignette highlighting a common clinical problem. Evidence supporting various strategies is then presented, followed by a review of formal guidelines, when they exist. The article ends with the authors' clinical recommendations.

N Engl J Med 2022;387:45-57.
DOI: 10.1056/NEJMc2116489

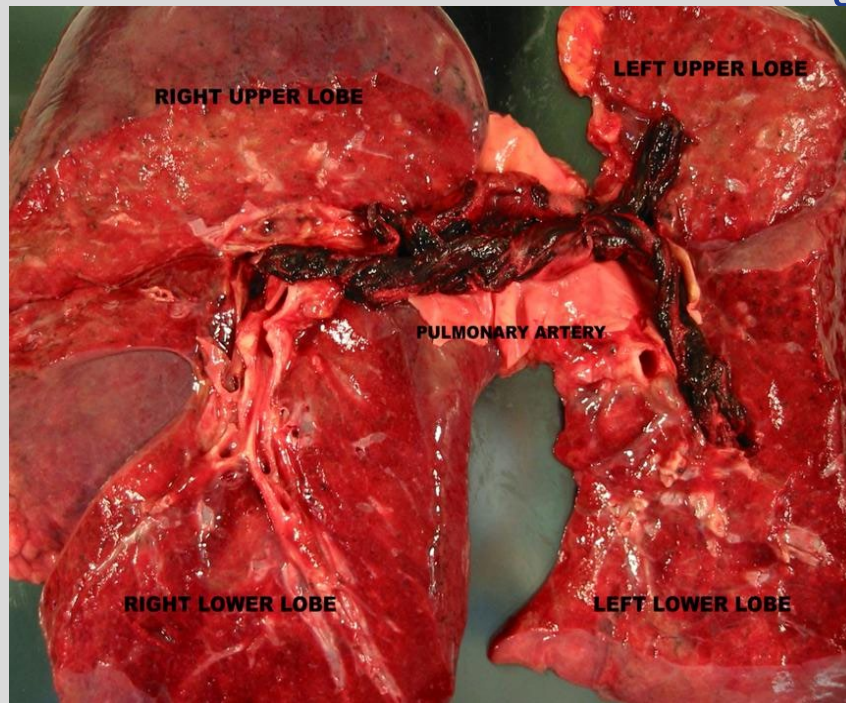
- **Deep Venous Thrombosis**
 - A blood clot that forms within a deep vein of the leg
- **Pulmonary Embolism**
 - Blockage of lung's main artery or one of its branches by a substance that has traveled from elsewhere in the body through the blood stream

- Between 100,000 and 300,000 VTE related deaths occur annually.
- US Surgeon general has declared that PE is most common **preventable cause of death** among hospitalized patients.

Saddle Pulmonary Embolism

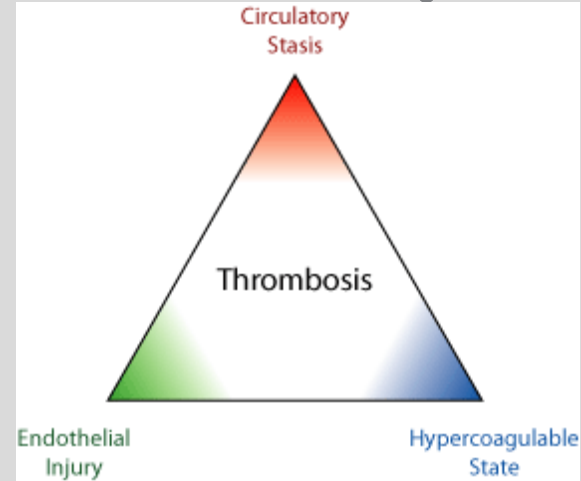
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Virchow's Triad

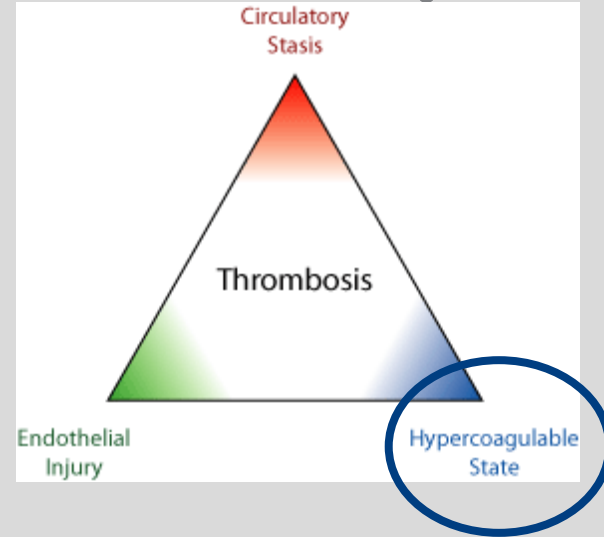
- Hypercoagulation
- Vascular Damage
- Venous Stasis





Virchow's Triad:

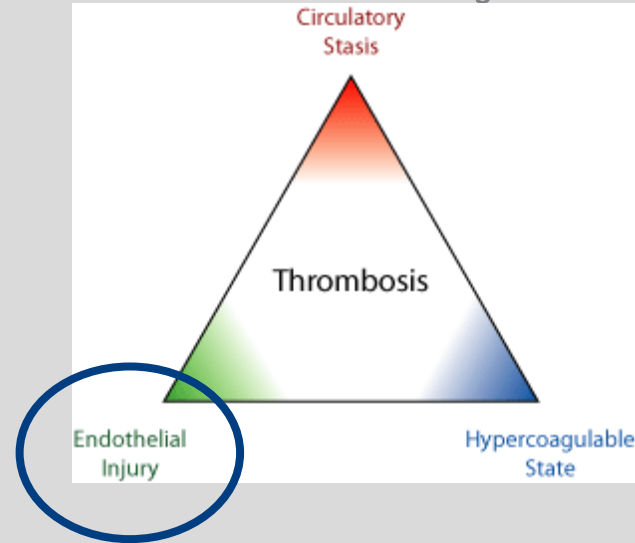
- **Hypercoagulation**
 - Inherited or acquired abnormalities of:
 - Coagulation System
 - Fibrinolytic System



Virchow's Triad:

• Vascular Damage

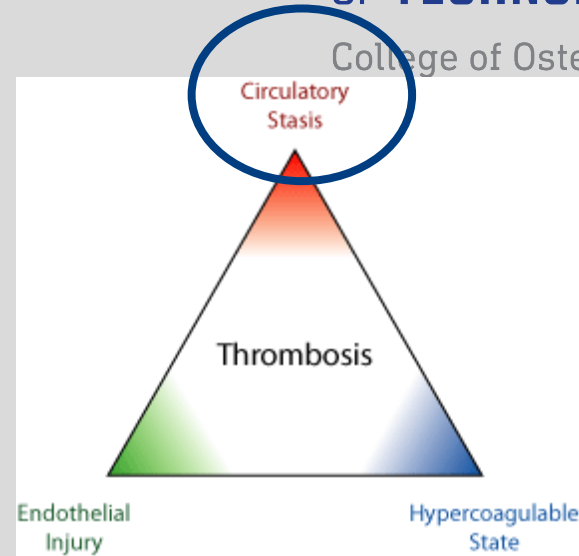
- Damaged vascular tissue results in release of tissue factors
- Subsequent activation of coagulation occurs



Virchow's Triad:

• Venous Stasis

- Activated coagulation factors remain undiluted and in contact with vascular endothelium
- Stasis also results in shielding from circulating inhibitors of coagulation
- Small thrombi are not washed away, viscosity increases



Clinical Risk Factors for Thrombus

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Immobility

History of previous thromboembolic disease

Use of estrogen

Pregnancy

Neoplasia

Congestive heart failure

Deficiency of Inherited Factors

Acquired Factors

Clinical Risk Factors:

Immobility

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- **Postoperative State**
 - Pelvic Surgery
 - Abdominal Surgery
 - Orthopedic Surgery
 - Hip fracture and replacement
 - Knee Replacement
- **Trauma**
- **Malignancy**
- **Cardiopulmonary disease**
 - Acute MI
 - Congestive Heart Failure
- **Obesity**

Clinical Risk Factors: Hx of Thromboembolic Disease

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History of previous thromboembolic disease

- This is especially important when exposed to surgery and immobilization
- Together these significantly increase risk of thromboembolism.



Clinical Risk Factors: Use of Estrogen

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Use of Estrogen

- Oral Contraceptive Use
 - Related to estrogen (not progesterone) dose
 - Risk up to 6x higher than baseline
 - ***Increased risk in smokers***
- Other estrogen
 - Large doses used in fertility
 - Estrogen used as adjunct in treatment of malignancy
 - Post-menopausal estrogen replacement therapy
 - This may possibly double risk



Clinical Risk Factors:

Pregnancy

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Pregnancy

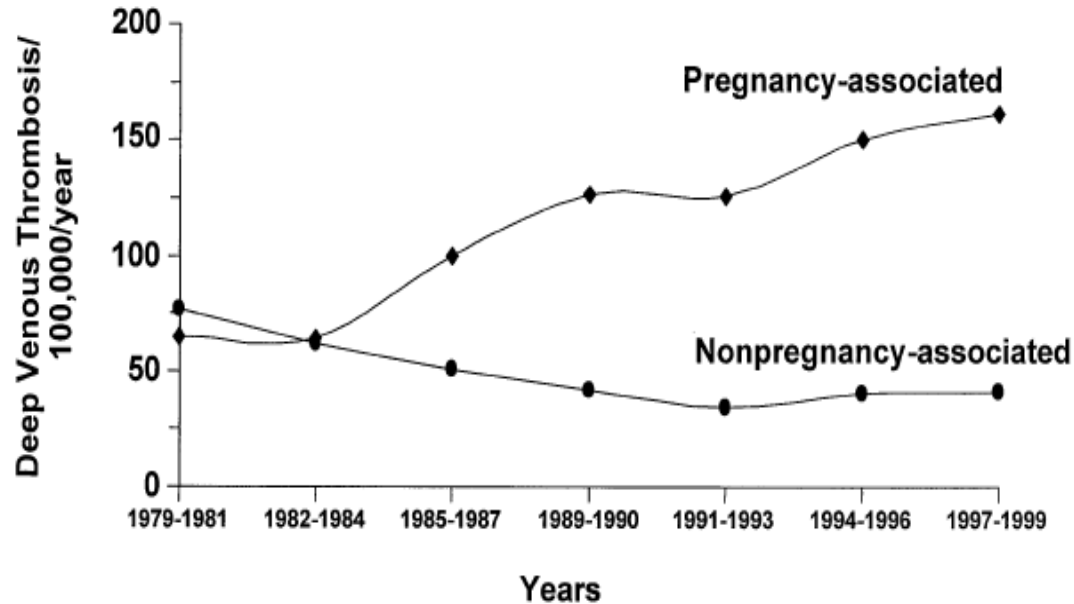
- Occurs in late stages and postpartum period
- Incidence 7x greater than in age-matched non-pregnant
- 2/3rds occur in immediate postpartum period
 - Lower extremity and pelvic venous stasis (due to uterine enlargement)
 - Venous dilatation, thrombocytosis, increased platelet adhesiveness
 - Release of tissue thromboplastin at time of placental separation
- ***Incidence of fatal PE after C-section is 9X greater than vaginal***

DVT Trends over 21 Year Period

abstracted from National Hospital Discharge Database

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Stein PD, et al. Am J Med 2004;117:121-125



Clinical Risk Factors: Neoplasia

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Neoplasia

- Described more than 100 years ago by Trousseau
- Greatest with neoplasms of lung, pancreas, breast, GI tract, and GU Tract
- Associated with procoagulant material (factor X activation)
- DVT may be a presenting manifestation of otherwise occult cancer.
- **Trousseau Sign of Malignancy is a migratory thrombophlebitis**



Clinical Risk Factors: **Inherited Factors**

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Inherited Factors

- Antithrombin III Deficiency
- Protein C Deficiency
- Protein S Deficiency
- Deficiency of plasminogen activators
- Factor V Leiden Mutation
 - G1691A mutation in Factor V gene
- G20210A mutation in 3' untranslated region of prothrombin gene



Clinical Risk Factors: Inherited Factors

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Factor V Leiden (G16191A mutation)

- Inherited thrombophilic defect
- Activated protein C (APC) resistance
- ***Most common cause of inherited thrombophilias***

G20210A Mutation in 3' untranslated region of prothrombin gene

- Results in 25% increase in prothrombin levels
- Occurs in 5% of patients

Deep Vein Thrombosis

***Most venous thrombi are silent and
produce no Symptoms***

- May be confined to intramuscular veins of calf
- may undergo spontaneous lysis
- some extend into popliteal and more proximal veins

Most symptomatic pulmonary emboli and ***virtually all fatal emboli*** arise from thrombi in *proximal veins of leg*

- *Iliac*
- *Popliteal*
- *Femoral*

50% of patients with symptomatic proximal vein thrombosis have silent pulmonary emboli



Venous Thrombosis:

Clinical Features

Most venous thrombi produce no symptoms

- Typical clinical features are non-specific:
 - Localized swelling
 - Tenderness, redness, edema
 - **Homan's Sign**
- **70%** of ambulatory patients clinically suspected to have venous thrombosis **will have another cause:**
 - Ruptured baker's cyst
 - cellulitis
 - muscle tear or hematoma
- Of 30% of patients with venous thrombosis
 - 85% have proximal DVT; remainder confined to calf



Deep Vein Thrombosis:

Clinical Features

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Deep Vein Thrombosis:

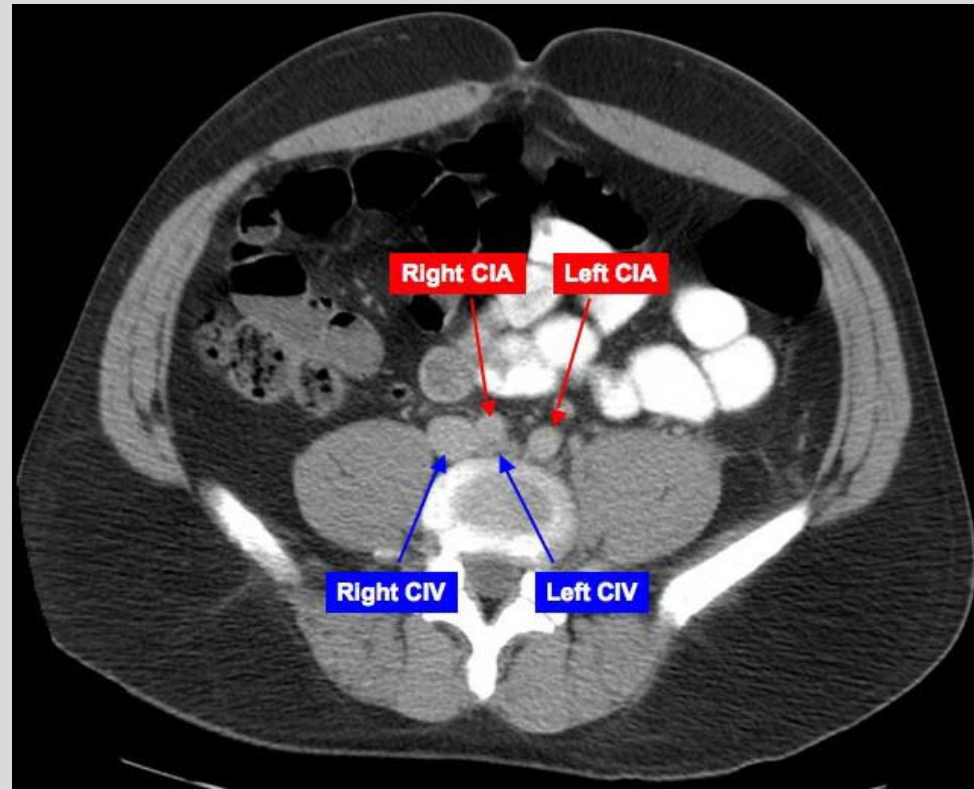
Clinical Features

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May-Thurner
Syndrome

Compressive
effects of the
Right iliac
artery
crossing
the Left iliac
vein





Deep Vein Thrombosis: Fate

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Resolution

- Fibrinolysis
 - Rapid dissolution of thrombus
 - Hours to Days
- Organization
 - Incorporation of residual thrombus into venous wall
 - Small zone of fibrotic tissue
 - Incompetence of valve may result causing postphlebitic syndrome
- **Embolization**
 - Risk greatest early when clot is friable
 - Risk *decreases* as clot organizes



Deep Vein Thrombosis: Fate

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When the valve becomes incompetent:



Post Phlebotic Syndrome

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Pulmonary Embolism

I ***ALWAYS*** think of PE

Always...

Always....



Pulmonary Embolism

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- Occurs when obstruction of pulmonary arteries by emboli affects lung tissue, the pulmonary circulation, as well as right and left heart function
- Degree of cardiopulmonary compromise correlates with:
 - Extent of embolic occlusion
 - Degree of preexisting cardiac and pulmonary disease

Clinical Features: Pulmonary Embolism

- Usually non specific
- 50% of patients suspected to have diagnosis are confirmed
- Presentation usually subtle
 - When suspected we must confirm the PE as well as DVT
- Clinical picture determined by
 - Pre-embolic cardiopulmonary status
 - Extent of embolization
- Cause of up to 10% of hospital deaths
 - Most go unrecognized



Pulmonary Embolism: Symptoms

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Dyspnea

- Most common symptom

Chest Pain

- Common, usually pleuritic

Anxiety

Syncope and near-syncope

Cough

Hemoptysis

- Less common

Pulmonary Embolism: Physical Exam

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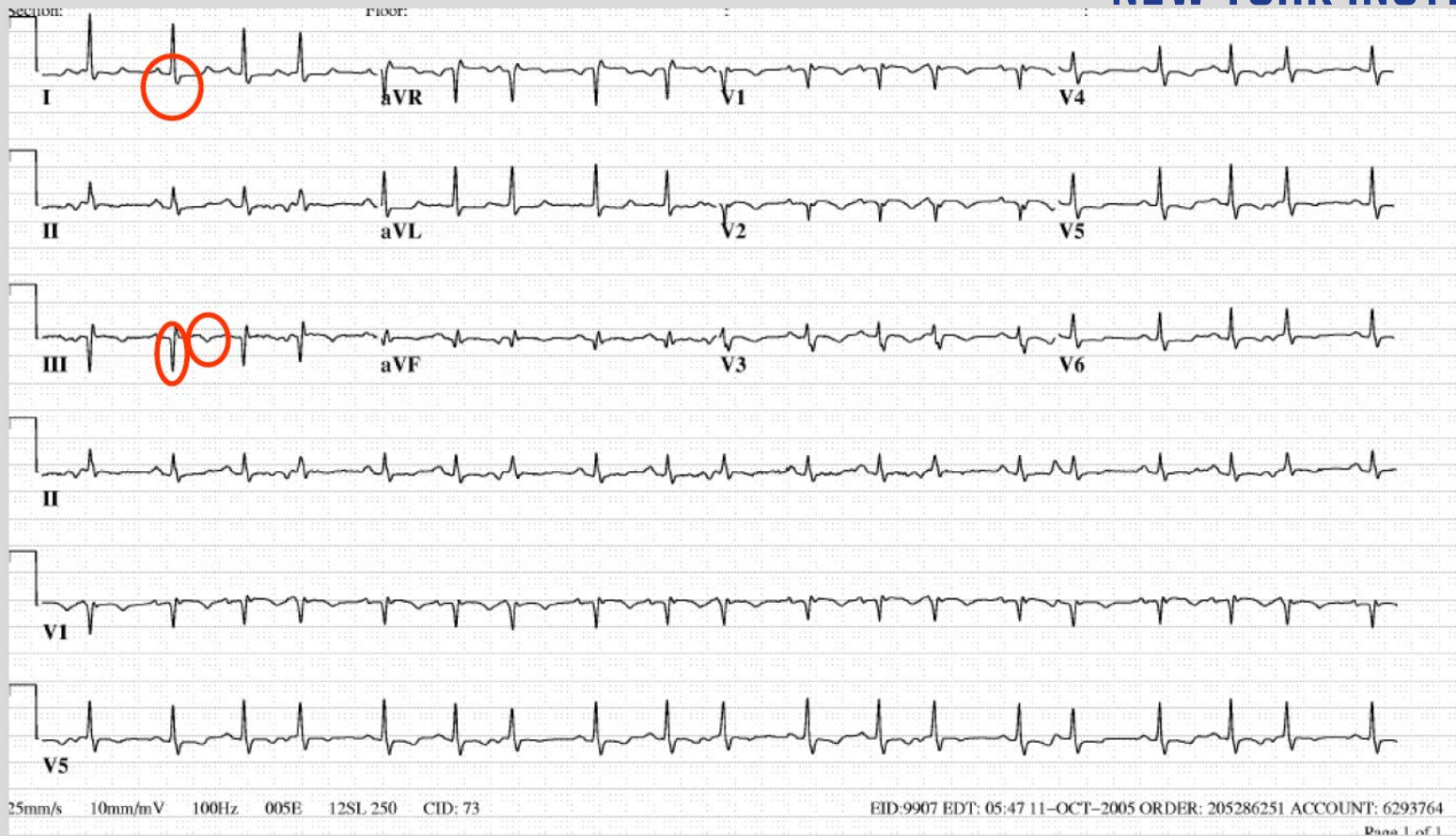
- Tachypnea- Consistent finding
- Tachycardia- up to 40% of patients, transient
- Fever- up to 40% of patients experience **low grade fevers** (100-101F)
- Focal Crackles
- Wheezing and cyanosis are rare

Pulmonary Embolism

Electrocardiogram

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- Generally shows non-specific alterations
- Frequently normal or sinus tachycardia
- Changes in S-T segment and T wave inversion common
- $S_1Q_3T_3$ pattern and *right axis deviation* usually indicate acute right-sided compromise
- EKG most useful for ruling out cardiac causes of patient's signs and symptoms
 - Acute myocardial infarction
 - Acute pericarditis



Pulmonary Infarction

- Rarely occurs, potentially serious
- Uncommon because lung receives multiple oxygen supply:
 - Bronchial circulation
 - Pulmonary circulation
 - Alveolar Oxygen
- More common with left ventricular failure, mitral stenosis, and COPD

Pulmonary Infarction

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Fate of Pulmonary Emboli

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Fragmentation

Dissolution

Organization

**Always think PE when the
patient has even a slight
possibility that a PE has
formed!**

**Death is preventable because
PE is treatable!**

- **When PE is being considered we must think of the presence of DVT as well**
- **Wells Criteria can help identify likelihood of PE based on existing risk factors and symptoms**



Pulmonary Embolism: Dx

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Modified Wells Criteria:

• Clinical symptoms of DVT	3
• Other diagnosis less likely than PE	3
• Heart rate >100	1.5
• Immobilization or surgery < 4 weeks	1.5
• Previous DVT/PE	1.5
• Hemoptysis	1
• Malignancy	1

Clinical Probability Assessment

• High	> 6
• Moderate	2 – 6
• Low	< 2



Diagnosing DVT

- *Venography* is the Reference (Gold) standard
- *Venous ultrasonography* is Diagnostic test of choice (2019)
 - Highly accurate detection of proximal vein thrombosis (95% sensitivity/95% specificity)
- **D-Dimer Testing**
 - Negative predictive value = 98.4%
 - Rules out a DVT with great certainty

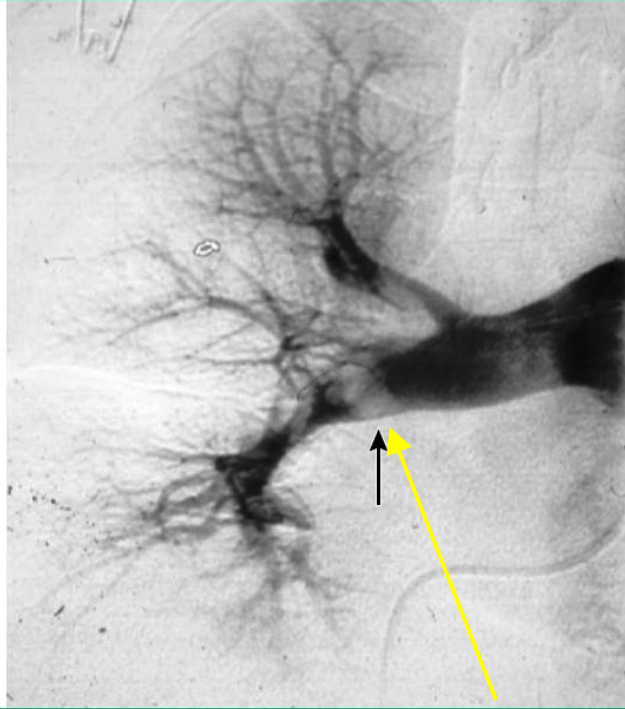
D-dimer level <500 ng/mL measured by ELISA is sufficient to exclude PE unless the pretest probability of PE is high

Pulmonary Embolism: Dx

- *Pulmonary Angiogram* is the Reference standard (Gold) for establishing presence or absence of pulmonary emboli
 - Findings will show an abrupt cutoff of flow in a major vessel or a filling defect
 - Normal angiogram virtually excludes the diagnosis of pulmonary embolism
- But up to 2% mortality and 5% morbidity
 - So rarely utilized

**Pulmonary
angiogram
demonstrates
non-filling of
pulmonary
arterial
branches in
the right lung,
an indication
of occlusion**

Catheter-based angiogram of a pulmonary embolus

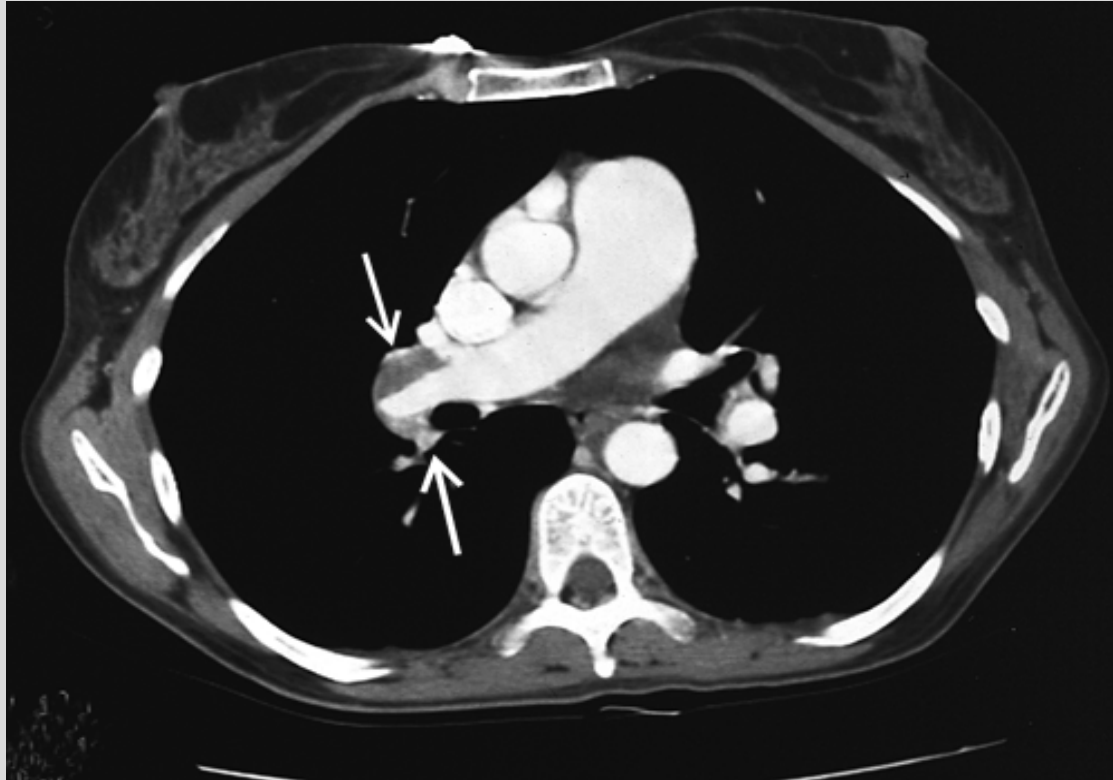


Pulmonary embolus. Frontal image from a selective right pulmonary artery angiogram shows a filling defect (arrow).

UpToDate®

Pulmonary Embolism: Dx

- CT Scan IV Contrast Angiography is the **Diagnostic test of choice**
- Sensitivity ~90% for emboli in segmental or larger pulmonary arteries.
- *Valuable to exclude other diagnoses:*
 - Pneumonia
 - Lung Cancer
 - Pneumothorax



Spiral CT Scan showing Central Emboli

Other testing modality:

Magnetic Resonance Angiography

- Shows promise as noninvasive method of diagnosing pulmonary embolism without need for radiation or iodinated contrast material
- Also has value in diagnosis of DVT, can be used to scan legs and lungs for clot at the same time

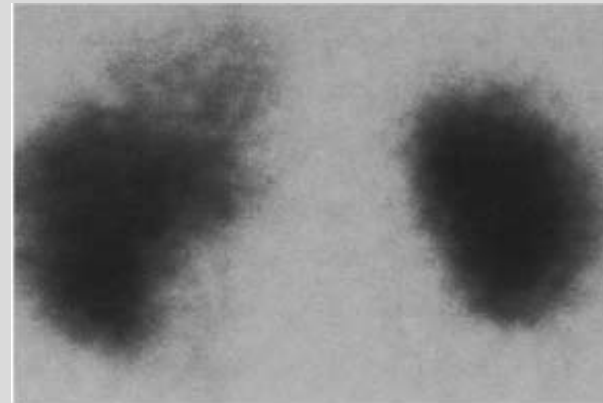
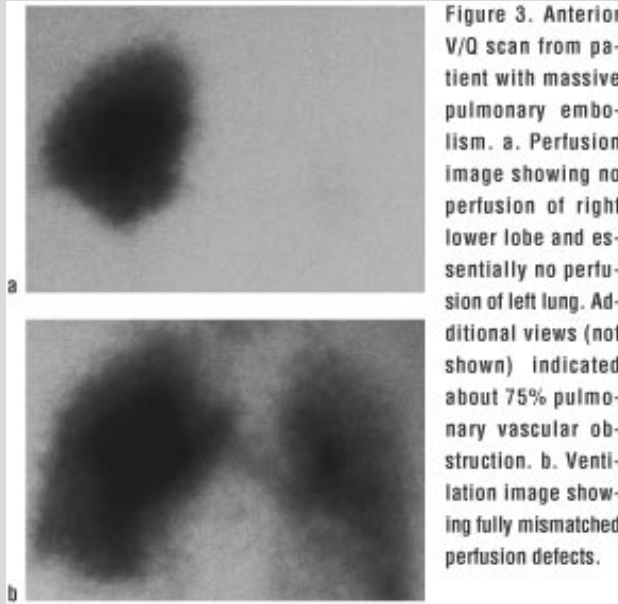


Other Testing Modalities:

• Ventilation-Perfusion (V/Q) Lung Scanning

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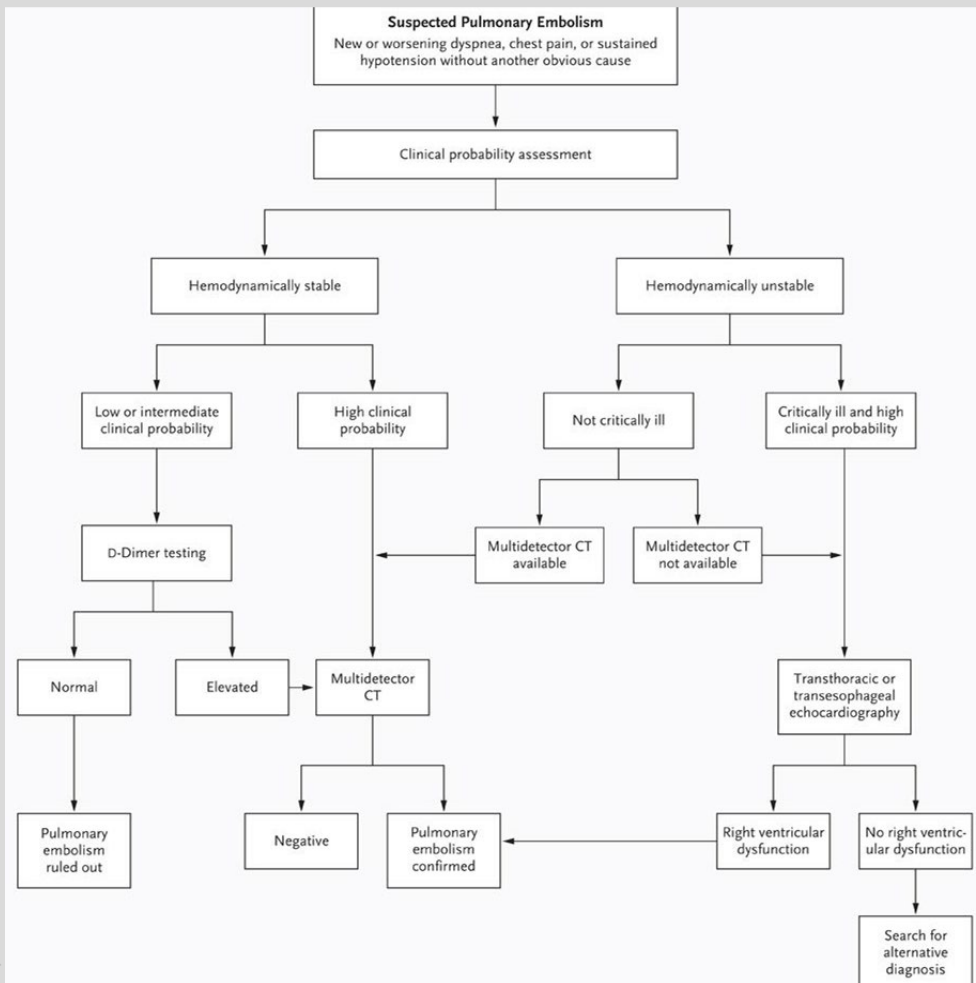


BEFORE

AFTER

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Prevention of Venous Thromboembolism

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- Prophylaxis in high-risk patients will reduce incidence of DVT significantly.
- Pharmacologic Methods (modulate activation of blood coagulation)
 - Heparin, warfarin, Factor Xa inhibitors, etc
- Mechanical Methods (prevent or reduce stasis)
 - Intermittent pneumatic compression
 - Graded compression stockings



Prevention of Venous Thromboembolism

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- **General Surgery and Medicine**
 - Low-dose heparin
 - **Reduces incidence of VTE by 50% to 70%**
 - Pneumatic compression if contraindicated
- **Hip Surgery**
 - **LMWH or oral anticoagulants, including Factor Xa inhibitors**
 - Provide effective prophylaxis for VTE
 - Aspirin somewhat effective, but not recommended
- **Major knee surgery**
 - **LMWH** and pneumatic compression effective
 - **LMWH treatment of choice**
 - Factor Xa inhibitors effective
- **Other surgery**
 - Pneumatic compression effective and recommended



Venous Thromboembolism: Tx

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Low Molecular Weight Heparin (LMWH)

- Effective in prevention and treatment of venous thrombosis and pulmonary embolism
- Low incidence of side effects such as heparin-induced thrombocytopenia(HIT) and osteopenia
- **This is *treatment of choice***

Other treatments

- IV Heparin gtt
- Other LMWH
- Warfarin (Coumadin)
- NOACs

Oral Anticoagulants: NOAC

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- Factor Xa inhibitors
 - Rivaroxaban (Xarelto)
 - Apixaban (Eliquis)
- Direct Thrombin inhibitors
 - Dabigatran (Pradaxa)
- No lab monitoring
 - Peak efficacy 1-4 hours
- **Antidotes?**
 - **Praxbind**
 - **Andexxa**

Anticoagulants: Complications

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- **Complications of therapy primarily related to bleeding from excessive anticoagulation**
- Incidence of bleeding increases with age and duration and intensity of treatment

Coumadin:

- Crosses placenta and can cause spontaneous abortion and teratogenicity
- Large number of drug interactions
 - Protein-binding
 - Metabolism
- Skin necrosis
 - Related to effect of warfarin on protein C





Submassive PE vs Massive PE

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Submassive PE- RV Dysfunction
with SBP >90 mmHg

Massive PE- Hemodynamic
compromise with SBP <90, Shock

Pathophysiology of Major Pulmonary Embolism

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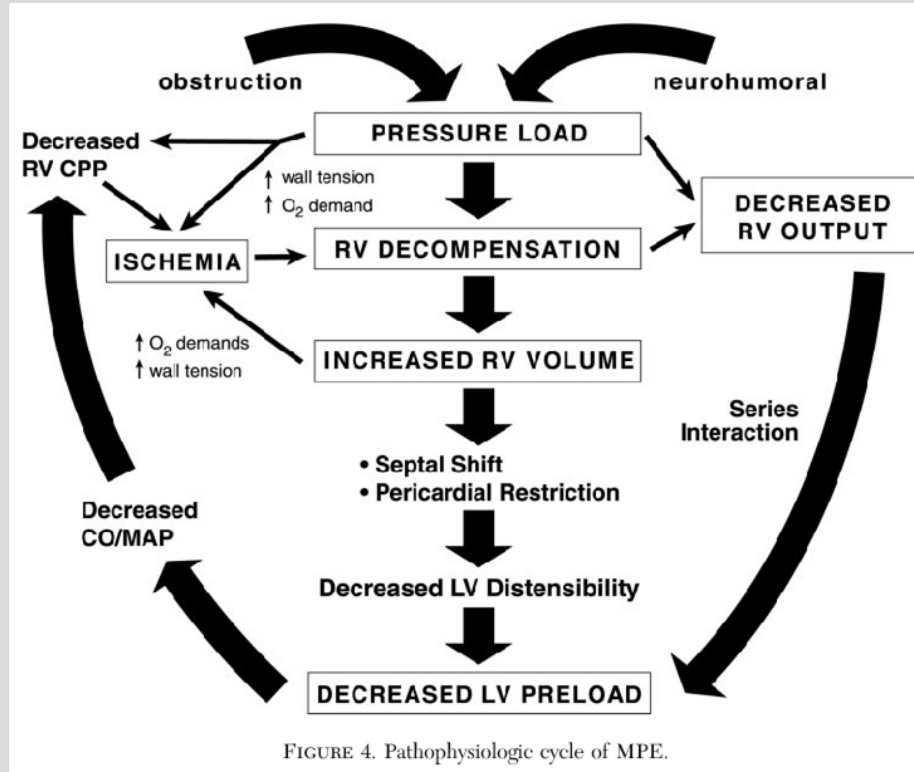


FIGURE 4. Pathophysiologic cycle of MPE.



Thrombolytic Therapy

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- Anticoagulant therapy reduces likelihood of recurrent thromboembolic disease *but it does NOT lyse the clot!*
- With a massive PE there will be more rapid and direct compromise of cardiopulmonary function, *to fix this we need to lyse the clot*

Thrombolytic Therapy cont'd.

- Pharmacologic thrombolysis produced by **plasminogen activators**
 - Streptokinase
 - Tissue plasminogen activator (tPA)
 - Urokinase
- **These compounds convert proenzyme plasminogen to fibrinolytic-enzyme *plasmin***



The **American College of Chest Physicians** produced guidelines on use of thrombolytic therapy in PE:

- For most patients with PE, ***we recommend clinicians not use systemic thrombolytic therapy (Grade 1A)***
- For patients who are ***hemodynamically unstable*** we ***suggest use of thrombolytic therapy (Grade 2B)***



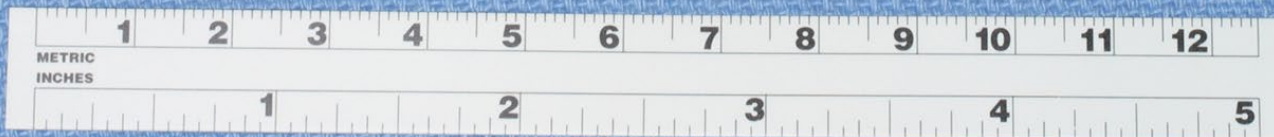
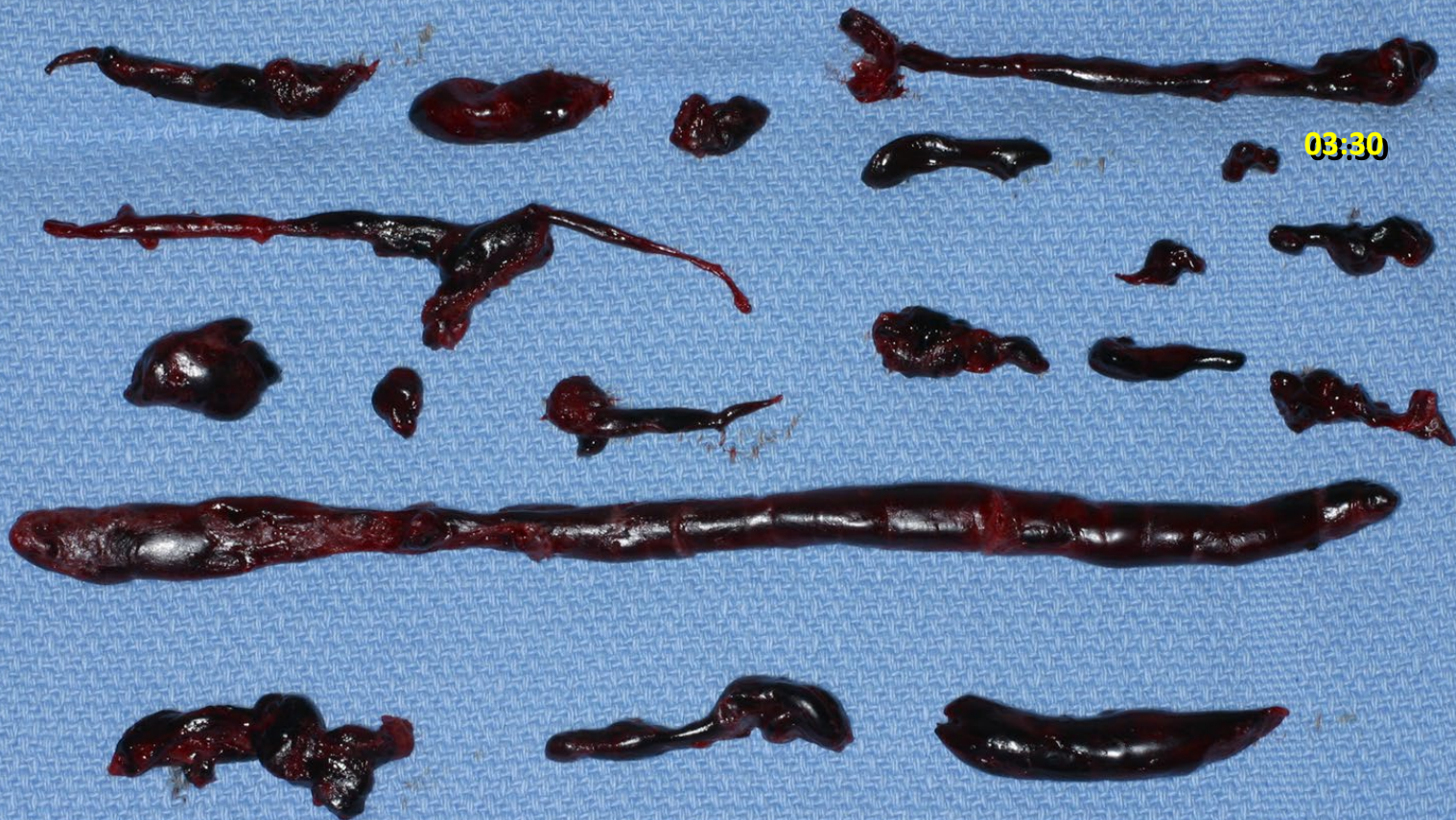
Other Treatment Approaches

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- **Surgical thrombectomy**
 - Used when there is a contraindication to thrombolysis
 - Failure of thrombolytics
 - Large RA/RV Thrombus
- **Vena Cava Interruption**
 - Used when anticoagulants cannot be administered, or there are complications of anticoagulant therapy like hemorrhage
 - Also used if there is recurrent PE *despite* appropriate and adequate anticoagulation
 - High Risk patients with large DVT
 - Allows patency of IVC but prevents clots from passing

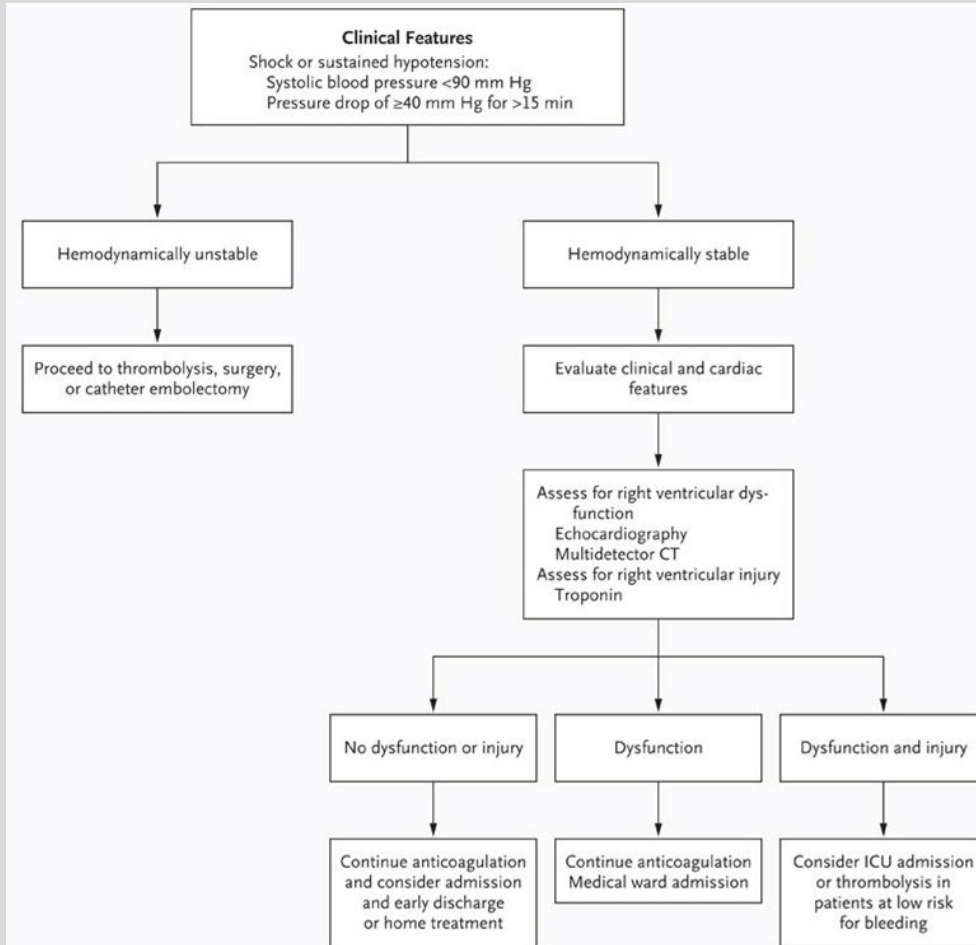
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Some points of interest

- Virchow's Triad and other clinical risk factors will point you to a diagnosis of PE
- Diagnostic tests of choice vs. gold standards
- PE majority of the time is caused by a DVT, thus when exploring a diagnosis of PE you should explore for possible DVT too
- Submassive PE does not necessarily require thrombolysis, anticoagulation may be enough
- Massive PE will require thrombolysis

I ***ALWAYS*** think of PE

Always...

Always....

Thank you! Any questions?